EMPIRICAL PAPER I

COMPENSATED CONTINUITY IN CANADIAN HOSPITALS

A First Empirical Test of the SignalRupture Structural Grammar

SignalRupture (SR)

Post-Gauntlet Empirical Series — August 2026

Abstract

This paper conducts the first formal empirical probe of SignalRupture (SR) after its reformulation as a developing structural metatheory and cross-domain structural grammar. The test is intentionally narrow. It does not attempt to prove societal collapse, validate the full SR architecture, or establish Compensatory Inversion as a theorem. It tests the first measurable SR proposition: observable continuity can coexist with a change in the substrate carrying that continuity when extraordinary compensation expands under unresolved capacity pressure. Canadian hospital staffing provides an unusually direct case because regular hours, overtime hours, and purchased staffing hours are reported longitudinally by the Canadian Institute for Health Information (CIHI). From 2019–2020 to 2023–2024, purchased staffing hours increased 126% nationally in the available hospital series and overtime hours increased 100%. Rural/remote purchased hours increased 261%, compared with 98% in urban hospitals. At the same time, regular hours remained the dominant staffing substrate and hospitals continued delivering large volumes of care, while access pressure remained visible in subsequent emergency-department data. In 2024–2025, CIHI recorded 16.1 million emergency-department visits; 7.7% ended with patients leaving before physician assessment, and overall ED waits were higher than in 2018–2019. The observed configuration is consistent with the SR state of Compensated Continuity: continuity is preserved while the composition of capacity shifts toward compensatory labour. However, the evidence does not yet establish a causal Q–D deficit, does not demonstrate that overtime or purchased staffing is necessarily depleting, and does not confirm Compensatory Inversion. The result is therefore classified as preliminary empirical support for the structural grammar's diagnostic distinction between observable function and durable structural sufficiency, with the predictive theorem remaining prospective.

Keywords: SignalRupture; structural grammar; compensated continuity; healthcare; overtime; agency staffing; hospital capacity; emergency departments; Canada; empirical validation.

1. Research Question

The post-gauntlet SR program asks whether a small relational grammar can be operationalized across different systems without redefining its logical structure after outcomes are known. The present study addresses only the healthcare instantiation. Its primary question is:

Can Canadian hospitals preserve substantial observable service continuity while the staffing substrate shifts toward extraordinary compensation?

A positive answer would not establish that SR is unique, causal, or superior to resilience engineering. It would establish that the proposed grammatical distinction is empirically observable in the first domain selected for formal testing.

2. Frozen Grammar Applied to Healthcare

SR role	Healthcare operationalization	Status in this paper
Q — effective requirement	Workforce/service requirement implied by realized demand and access obligations	Partially observed; no single scalar Q estimated
D — durable capacity	Regular sustainable staffed capacity	Observed primarily through regular hospital hours
C_S — sustainable flexibility	Historically normal renewable overtime/float/casual flexibility	Not cleanly separable with current aggregate series
C_E — extraordinary compensation	Overtime above sustainable norm and purchased/agency staffing	Purchased staffing directly observed; overtime observed but extraordinary share not fully identified
F_volume — observable function	Delivered hospital/ED activity	Observed

F_access — functional access	Waits, leaving before assessment, benchmark performance	Observed in later CIHI access series
HB — human buffer	Retention, absence, vacancy duration, replenishment	Not sufficiently measured here

The frozen protocol explicitly prohibits treating every overtime hour as pathological. The empirical series therefore measures compensatory dependence, while the stronger classification C_E remains provisional until a sustainable baseline is estimated.

3. Hypotheses

H1 — Compensation Expansion.

Under staffing pressure, C_t should increase relative to ordinary staffing.

H2 — Compensated Continuity.

$$C_t \uparrow \text{ while substantial } F_{volume} \text{ is maintained.}$$

H3 — Surface/Substrate Divergence.

Stable or substantial F_{volume} does not imply that staffing composition $D/(D+C)$ is unchanged.

H4 — Compensatory Inversion.

$$\Delta C_E \leq 0 \wedge G_C > 0 \wedge \Delta F < 0$$

H4 is not expected to be confirmable from the current aggregate dataset. It is included because the first empirical paper must distinguish what the evidence can test now from what remains prospective.

4. Data

The primary staffing source is CIHI's updated 2024 health-workforce staffing and service-delivery series. CIHI reports hospital regular, overtime, and purchased hours by urban versus rural/remote location from 2019–2020 through 2023–2024. The published series has coverage limitations: data exclude Quebec and Nunavut; 2022–2023 and 2023–2024 exclude Saskatchewan; and purchased-hours data for 2023–2024 exclude

Nova Scotia. These exclusions prevent the series from being interpreted as a perfectly complete national census and are retained as a central limitation.

For functional access, the study uses CIHI's 2026 emergency-department reporting. CIHI reports 16.1 million ED visits in 2024–2025 in NACRS, representing about 89% of Canadian ED visits. Overall ED waits increased between 2018–2019 and 2024–2025, and 1.2 million visits (7.7%) ended with patients leaving before physician assessment. These access data are not contemporaneously matched to the staffing panel at hospital level and therefore provide contextual outcome evidence rather than a causal linkage.

5. Descriptive Staffing Results

Setting	2019–20 regular %	2023–24 regular %	2019–20 overtime %	2023–24 overtime %	2019–20 purchased %	2023–24 purchased %
Rural/remote	95.0	91.2	3.9	5.5	1.0	3.4
Urban	96.9	94.4	2.5	4.7	0.5	0.9

The composition shift is unambiguous. In rural/remote hospitals, regular hours declined from 95.0% to 91.2% of worked hours while overtime rose from 3.9% to 5.5% and purchased hours rose from 1.0% to 3.4%. In urban hospitals, regular hours declined from 96.9% to 94.4%, overtime rose from 2.5% to 4.7%, and purchased hours rose from 0.5% to 0.9%.

CIHI reports more than 7.8 million purchased staffing hours in 2023–2024, equivalent to 4,010 FTEs, a 126% increase from 2019–2020. Providers also worked almost 32 million overtime hours, equivalent to 16,397 FTEs, a 100% increase from 2019–2020. CIHI explicitly describes purchased staffing and overtime as mechanisms used to fill urgent staffing gaps.

6. Rural/Remote Exposure Test

The rural/remote comparison provides a simple exposure contrast. Purchased hours increased 261% in rural/remote hospitals between 2019–2020 and 2023–2024, compared with 98% in urban hospitals. In 2023–2024, purchased hours accounted for more than 3%

of worked hours in rural/remote hospitals and less than 1% in urban hospitals. CIHI links the rural/remote pattern to greater recruitment and retention challenges.

$$\text{Greater recruitment/retention constraint} \rightarrow \text{greater purchased-staffing dependence}$$

This directional result is consistent with H1. It is not a causal estimate because geography is associated with many other determinants of staffing and service delivery.

7. Overtime Plateau Signal

Year	Rural overtime %	Urban overtime %	Rural purchased %	Urban purchased %
2019–20	3.9	2.5	1.0	0.5
2020–21	3.8	2.9	1.2	0.5
2021–22	5.0	4.1	1.7	0.7
2022–23	5.6	5.0	2.7	0.9
2023–24	5.5	4.7	3.4	0.9

Overtime shares stopped rising in the final observed year: rural/remote overtime moved from 5.6% to 5.5% and urban overtime from 5.0% to 4.7%. This cannot be classified as Compensatory Inversion because purchased staffing continued to rise in rural/remote hospitals and because the underlying Q–D gap and matched functional outcomes are not yet estimated. The observation is therefore registered only as a future signal requiring follow-up.

8. Functional Continuity

The structural-grammar claim does not require that every functional measure remain healthy. It requires that substantial visible function can continue while the substrate supporting it changes. Canadian hospitals plainly continued operating during the period of compensation expansion. The later ED series records 16.1 million visits in 2024–2025, with more than 14.8 million completed visits in the CIHI dataset. This is incompatible with a literal interpretation that the hospital system had ceased functioning.

$$F\_volume\ substantial \wedge compensation\ dependence \uparrow$$

This is the empirical form of Compensated Continuity. It is deliberately weaker than a claim of adequate performance.

9. Functional Deterioration in Access

Continuity and access are not equivalent. CIHI's 2026 ED analysis reports that overall waits increased from 2018–2019 to 2024–2025. Half of ED patients in 2024–2025 waited just under two hours for initial physician assessment, while one in ten waited more than five hours. CIHI also reports 1.2 million visits, or 7.7%, where patients left before being seen and assessed by a physician.

The combined pattern is therefore not simply "function stable." It is better represented as:

$$F\_volume\ maintained/substantial\ while\ F\_access\ shows\ pressure$$

This distinction is important for SR because a headline volume measure can remain large while latency and access deteriorate. It is also important methodologically: the grammar cannot define F as whichever outcome produces confirmation. Volume and access must remain separate predeclared dimensions.

10. Does the Evidence Confirm Compensated Continuity?

Requirement	Observed?	Assessment
Compensatory staffing rises	Yes	Strong
Regular staffing remains dominant but its share falls	Yes	Strong
Substantial service activity continues	Yes	Strong
Access pressure exists	Yes	Strong contextual evidence
Direct Q–D deficit measured	No	Incomplete
C_S separated from C_E	No	Incomplete
Matched staffing → outcome panel	No	Incomplete

Causal buffering effect estimated	No	Not established

The appropriate result is therefore not “SR proven.” The observed data support the diagnostic configuration that motivated the grammar: visible continuity can coexist with increasing reliance on compensatory capacity. The stronger causal proposition—that compensation is what preserved function relative to the counterfactual—is not identified by this descriptive design.

11. Alternative Explanations

Several alternatives can produce the observed staffing composition without implying depletion. Demand may have risen; overtime may reflect deliberate scheduling; agency staffing may represent efficient labor-market substitution; pandemic-era workforce changes may have altered preferences; funding changes may have allowed hospitals to purchase more labor; and changes in service mix may affect both staffing and waits. These alternatives do not necessarily contradict Compensated Continuity, but they constrain any claim that compensation is pathological or that it causes later deterioration.

The empirical value of SR therefore depends on the next-stage matched longitudinal model, not on the descriptive pattern alone.

12. Relationship to Established Science

The finding is compatible with organizational slack, resilience engineering, high-reliability research, workforce planning, and health-services research. SR does not claim that the use of overtime or temporary staff under pressure is novel. Its candidate contribution is the grammatical separation of durable capacity, sustainable flexibility, extraordinary compensation, and observable function, followed by a prospective test of whether compensation trajectory adds predictive information beyond conventional workforce and access indicators.

If conventional healthcare models predict future access and discontinuity equally well without the SR variables, this domain will not establish incremental metatheoretical value.

13. First Statistical Specification

The current paper is descriptive because publicly available aggregate series do not yet provide a sufficiently harmonized unit-level panel for the frozen causal/predictive test. The next empirical stage is specified in advance:

$$M0: F_{(i,t+h)} = \alpha + \beta 1F_{(i,t)} + \beta 2Q_{(i,t)} + \beta 3D_{(i,t)} + \gamma Z_{(i,t)} + u_i + \tau_t + \varepsilon_{(i,t)}$$

$$M1: M0 + \beta 4C_{E,(i,t)} + \beta 5\Delta C_{E,(i,t)} + \beta 6[G_{C,(i,t)} \times \Delta C_{E,(i,t)}]$$

The empirical question is whether M1 improves out-of-sample prediction of access latency, closures/diversions, backlog, vacancy persistence, or other predeclared outcomes relative to M0 and simple baselines such as overtime level alone.

14. What Would Falsify the Healthcare Grammar

- Falling overtime and purchased staffing accompanied by improving staffing durability and access: this is restoration, not exhaustion.
- Persistently high compensation with stable retention, absence, vacancies, cost, and access over long periods: the supposedly extraordinary buffer may be sustainable.
- No relationship between Q–D mismatch and compensation after appropriate demand and policy controls.
- No incremental predictive value from compensation trajectory beyond standard healthcare indicators.
- Repeated need to redefine overtime, purchased staffing, Q, D, or F after outcomes are known.

15. Empirical Classification

SR proposition	Result
P1: Observable continuity does not establish durable sufficiency	Supported as a diagnostic distinction
P2: Compensation can coexist with maintained function under pressure	Supported descriptively
P3: Compensation trajectory predicts future function	Indeterminate / not yet tested

P4: Compensatory Inversion precedes deterioration	Indeterminate / prospective
P5: Restoration reduces extraordinary dependence	Not tested in this paper
P6: Grammar transfers across domains	Requires the electricity and household studies

16. Why This Counts as an Empirical Paper

The paper does not merely restate theory. It freezes a domain translation, confronts it with observed longitudinal staffing data, identifies a configuration that is present, identifies components that are absent or undermeasured, and preserves prospective hypotheses that the current data cannot confirm. That distinction is essential. Empirical science includes partial confirmation, indeterminacy, and failure; it does not require every hypothesis to succeed in the first dataset.

17. What This Paper Does Not Confirm

- It does not confirm SR as a universally valid metatheory.
- It does not establish that Canada is in healthcare collapse.
- It does not establish that overtime causes access deterioration.
- It does not establish that every purchased staffing hour is extraordinary or unsustainable.
- It does not validate Human Buffer as a scalar.
- It does not validate Compensatory Inversion.
- It does not demonstrate incremental predictive superiority over established healthcare models.

18. What This Paper Does Confirm

Within the limits of the available aggregate data, the Canadian hospital case confirms that the structural distinction SR asks researchers to measure is real and observable: the composition of the capacity supporting service continuity can change substantially while large-scale service activity continues. In the observed period, regular hours lost share while overtime and purchased staffing gained share, especially in rural/remote hospitals.

Later access data simultaneously show that continued operation should not be confused with absence of service pressure.

Observable continuity ≠ invariant staffing substrate

This is the first empirical foothold for SR as structural grammar. It is not yet the decisive predictive validation.

19. Prospective Commitment

The healthcare study now enters the frozen prospective phase. The 2027–2029 test will determine whether compensatory staffing dependence normalizes through restoration, persists as a sustainable equilibrium, or exhibits the Compensatory Inversion configuration. The classification may not be changed retrospectively to preserve SR.

Restoration: D/Q improves + C_E dependence ↓ + F improves/stabilizes

Candidate CI: C_E plateaus/falls + G_C remains positive + F deteriorates

20. Conclusion

Canadian hospital data provide preliminary empirical support for the first and weakest—but foundational—claim of the SignalRupture structural grammar: visible continuity can coexist with a materially changing compensatory substrate. Between 2019–2020 and 2023–2024, overtime and purchased staffing increased sharply, with particularly strong purchased-staffing growth in rural and remote hospitals. Hospitals nevertheless continued delivering large volumes of care. Subsequent ED evidence shows substantial continuing activity alongside worsening wait pressure and a meaningful share of patients leaving before physician assessment.

The result does not prove that extraordinary compensation caused continuity, nor that the human buffer is exhausted. It establishes that the surface/substrate distinction can be operationalized with real data and that the system cannot be characterized adequately by output volume alone. This is precisely the level at which the first empirical paper should succeed or fail.

The stronger SR claim remains ahead of the evidence. If the trajectory of extraordinary compensation prospectively improves prediction of functional deterioration after controlling for ordinary capacity, demand, and established sector indicators—and if the same logical structure survives tests in electricity and households—SR will have evidence for more than a descriptive vocabulary. Until then, the appropriate conclusion is narrower: the Canadian healthcare case supports Compensated Continuity as an empirically observable structural configuration and justifies the prospective test of the grammar.

References

Canadian Institute for Health Information. (2025, updated June 25, 2026). The state of the health workforce in Canada, 2024: Staffing and service delivery.

Canadian Institute for Health Information. (2025). Health workforce: Overtime and staffing challenges in hospitals.

Canadian Institute for Health Information. (2026). Emergency department wait times in Canada: Insights from a health system perspective.

Canadian Institute for Health Information. (2026). Wait times in Canada, 2026.

Bourgeois, L. J., III. (1981). On the measurement of organizational slack. Academy of Management Review, 6(1), 29–39.

Hollnagel, E., Woods, D. D., & Leveson, N. (Eds.). (2006). Resilience Engineering: Concepts and Precepts. Ashgate.

Meyer, J. W., & Rowan, B. (1977). Institutionalized organizations: Formal structure as myth and ceremony. American Journal of Sociology, 83(2), 340–363.

Woods, D. D. (2015). Four concepts for resilience and the implications for the future of resilience engineering. Reliability Engineering & System Safety, 141, 5–9.

Appendix A. CIHI Staffing Series Used

Fiscal year	Location	Regular hours	Overtime hours	Purchased hours
2019–20	Rural/remote	54,884,334	2,275,113	588,300
2020–21	Rural/remote	56,650,176	2,256,052	732,030

2021–22	Rural/remote	55,960,869	2,977,781	999,610
2022–23	Rural/remote	55,595,904	3,416,625	1,632,939
2023–24	Rural/remote	57,505,223	3,441,271	2,125,271
2019–20	Urban	521,720,374	13,549,519	2,879,267
2020–21	Urban	538,337,797	15,903,156	2,929,351
2021–22	Urban	546,592,063	23,280,540	3,820,859
2022–23	Urban	543,230,740	28,715,088	4,998,015
2023–24	Urban	579,186,153	28,531,980	5,695,030

Coverage notes: CIHI states that these staffing data exclude Quebec and Nunavut; 2022–2023 and 2023–2024 exclude Saskatchewan; purchased-hours data for 2023–2024 exclude Nova Scotia. FTE calculations in the source use normal earned hours defined by CIHI.

Appendix B. Frozen Score

Test	**August 2026 score**	**Reason**
Compensation expansion	Supported	Overtime and purchased hours increased materially.
Compensated continuity	Partially supported	Large-scale activity continued; causal buffering not identified.
Surface/substrate divergence	Supported descriptively	Staffing composition shifted while service continued.
Compensatory inversion	Indeterminate	Final-year overtime plateau insufficient; purchased staffing and matched outcomes prevent classification.
Predictive increment	Indeterminate	Requires matched longitudinal model and holdout testing.
Metatheory confirmation	Not claimed	Requires cross-domain replication and incremental value.